

Logistic Regression Based Fault Diagnosis of Induction Motor Using Current Signature Analysis

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Abstract—Fault diagnosis of induction motors is crucial for ensuring reliable operation and reducing maintenance costs in industrial systems. This paper presents a Logistic Regression-based approach for classifying motor health conditions using three-phase current signature data. Raw current signals are divided into fixed-length samples to enable both binary (healthy/unhealthy) and multiclass(14 classes) fault classification. Experimental results show that the proposed method achieves reliable diagnostic performance with low computational complexity, making it suitable for practical and real-time applications.

Index Terms—Induction Motor, Fault Diagnosis, Logistic Regression, Current Signature Analysis, Machine Learning

I. PROBLEM STATEMENT

Early and accurate detection of faults in induction motors is critical for improving system reliability, minimizing downtime, and preventing devastating failures. The objective of this work is to design a machine learning-based fault classification framework using Logistic Regression that can identify motor health conditions from three-phase current signals under different fault types and load conditions.

II. PROBLEM DESCRIPTION

The provided dataset consists of three-phase currents results collected from an induction motor operating under healthy and unhealthy conditions. The unhealthy conditions include bearing faults (inner-race and outer-race) with different faults and broken rotor bar (BRB) faults under multiple load levels. The problem is divided into two tasks: binary classification between healthy and unhealthy motor states, and multiclass classification involving 14 distinct motor health conditions.

III. DATASET PREPROCESSING

Each dataset file contains more than 100,000 samples per phase, recorded at a sampling rate of 10 kHz. To prepare the data for machine learning, the continuous current signals are divided into non-overlapping blocks of 1000 samples per channel. Each block is considered as one data sample and is labeled according to the motor's operating condition. All processed samples are saved in CSV format. Feature normalization is applied to maintain numerical stability, and the dataset is divided into training and testing sets.

IV. METHODOLOGY

Logistic Regression is selected due to its simplicity, interpretability, and efficiency. For binary classification, a standard Logistic Regression model is trained to distinguish between

healthy and unhealthy motor conditions. For multiclass classification, a one-vs-all strategy is used to extend Logistic Regression to handle 14 classes. Model performance is evaluated using evaluation metrics (accuracy, precision, recall, sensitivity, specificity, and F1-score).

V. PSEUDOCODE

- Acquire three-phase current signal data from the induction motor.
- Segment the continuous signals into blocks of 1000 samples.
- Assign motor health labels to each segmented block.
- Normalize the segmented data to ensure consistent feature scaling.
- Split the dataset into training and testing subsets.
- Train the Logistic Regression model for binary and multiclass classification.
- Predict motor fault classes for the test data.
- Evaluate model performance using evaluation metrics.

VI. FRAMEWORK

Data Acquisition → Preprocessing → Logistic Regression Training → Prediction → Performance Evaluation

VII. RESULTS AND DISCUSSION

The Logistic Regression model demonstrates effective performance in both binary and multiclass classification tasks. Binary classification achieves high accuracy in distinguishing healthy and unhealthy motor conditions. In the multiclass scenario, the model successfully differentiates among 14 fault classes, although performance slightly decreases due to increased class complexity. Overall, the results indicate that Logistic Regression is a suitable model for induction motor fault diagnosis using current signature data.

Binary Classification Accuracy: 0.5586844368013758

Fig. 1. Logistic Regression binary Classification Accuracy

VIII. LIMITATIONS

Although the proposed method works well, it has some limitations. Logistic Regression works best when the data can be separated in a linear way, so it may not perform well for complex and nonlinear fault patterns. In addition, using only time-domain features may not capture all fault characteristics

Accuracy : 0.5587
Precision : 0.0061
Recall (Sensitivity): 0.1132
Specificity : 0.5691
F1 Score : 0.0116

Fig. 2. Binary Classification Evaluation metrics

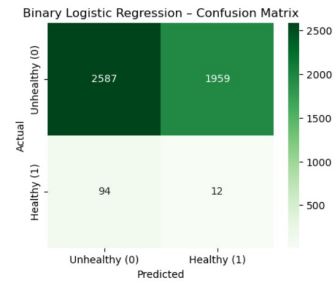


Fig. 3. Binary Logistic Regression-Confusion metrics

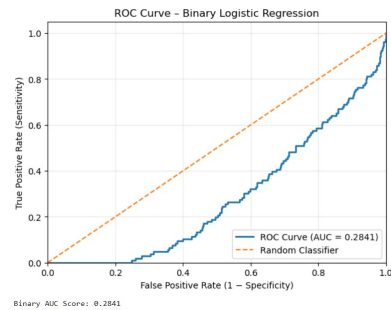


Fig. 4. ROC Curve-Binary Logistic Regression

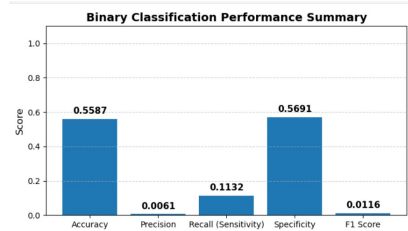


Fig. 5. Binary Classification Performance Summary



Fig. 6. Multi-Class Logistic Regression Accuracy

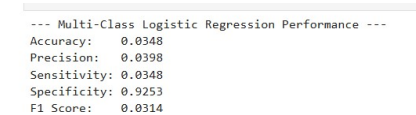


Fig. 7. Multi-Class Evaluation Metrics

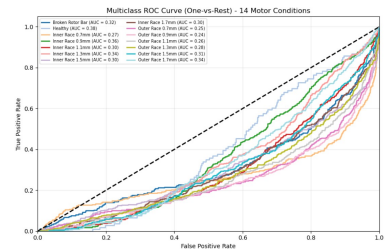


Fig. 8. Multi-Class ROC Curve

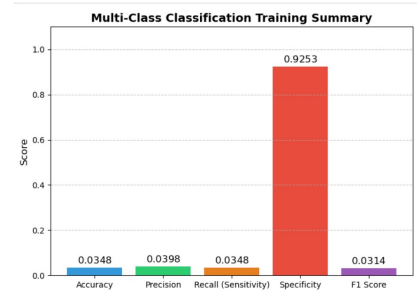


Fig. 9. Multi-Class Classification Summary

under different operating conditions. In future work, these limitations can be reduced by using more advanced feature extraction techniques and nonlinear classification methods to improve performance.

IX. CONCLUSION

This study presents a concise and effective Logistic Regression based approach for induction motor fault diagnosis using time-domain current signals. The proposed method provides reliable classification performance with low computational cost, making it suitable for real-time industrial applications.

REFERENCES

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